

Here's my personal **Engineering Subjects Resource Guide**, where I've broken down all the important subjects + the exact resources I personally used to crack internships, hackathons, and job offers + International High paying jobs/internships opportunities

This guide covers everything that actually matters for **placements** 📌

1. Data Structures & Algorithms (DSA)

Why it matters: It's the backbone of every tech interview — problem-solving, logic, and efficiency.

What to learn:

- Arrays, Strings, Stack, Queue, Linked List
- Trees, Graphs, Recursion, Sorting, Searching, Dynamic Programming

Type	Resource
YouTube	Striver's DSA Series (English) / CodeHelp (Hindi)
Practice	LeetCode 150 / CodeStudio / GFG "Top Interview Questions"
Tip	Focus on patterns, not memorizing solutions.
Most asked DSA interview questions	GFG and InterviewBit

2. Object-Oriented Programming (OOPs)

Why it matters: Every coding language & interview expects this.

What to learn:

- Classes, Objects, Inheritance, Polymorphism, Encapsulation, Abstraction
- Real-world examples using C++ / Java

Type	Resource
YouTube	Apna college , CodeHelp
Notes	Notes
Tip	Learn how to explain OOPs in simple language, it's often an HR-tech question.
Most asked interview questions	GFG , InterviewBit

3. Database Management System (DBMS)

Why it matters: It's a favorite in every company's technical round.

What to learn:

- ER Diagrams, Normalization, Joins, Transactions, Keys, SQL Queries

Type	Resource
YouTube	KnowledgeGate DBMS Series / Gate Smashers DBMS / CodeHelp
Practice	LeetCode SQL Section / GFG DBMS MCQs
Tip	Learn SQL hands-on using MySQL Workbench, not just theory.

4. Operating System (OS)

Why it matters: Commonly asked in every core interview — shows your CS fundamentals.

What to learn:

- Process Management, Deadlock, CPU Scheduling, Paging, Threads, Synchronization

Type	Resource
YouTube	CodeHelp
Notes	Notes

Most asked interview questions [InterviewBit](#), [Naukri](#)

5. Computer Networks (CN)

Why it matters: Helps in understanding how systems communicate — vital for backend/dev roles.
What to learn:

- OSI & TCP/IP Models, IP Addressing, Protocols, Routing, Switching

Type	Resource
YouTube	Neso Academy CN Series / Gate Smashers CN
Most asked interview questions	InterviewBit & GFG
Tip	Watch animations, it helps visualize how packets move across networks.

6. Development

Why it matters:

Because companies love candidates who can **build things**, not just solve problems on paper. Development proves your creativity, problem-solving, and end-to-end understanding of how tech works.

Learn any one of the technology:

- Web Development → HTML, CSS, JavaScript, React, Node.js, MongoDB
- App Development → Flutter or React Native
- AI/ML → Python, NumPy, Pandas, Scikit-learn, TensorFlow
- Blockchain → Smart Contracts, Solidity, Ethereum Basics

Type	Resource
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Tip Pick one domain, build **2 - 3 real-world projects**, and host them online, it'll make your resume stand out instantly.

Software Engineering Internship & New Grad Positions

[International Positions](#) (High paying jobs/internships)

This repository is a comprehensive list of Software Engineering jobs for college students in search of internships or new graduate positions. The positions are updated daily, and we prioritize jobs posted within the last 120 days.

How to Use This Guide

- Pick **one subject per week** → watch → summarize notes → revise weekly.
 - Once you finish these, 90% of your **placement-level technical prep** is done.
 - You can apply the same topics for **internship interviews, coding rounds, and projects**.
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Stay consistent, these subjects are your foundation.
Master them, and you'll fly through any **technical interview or coding test**

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